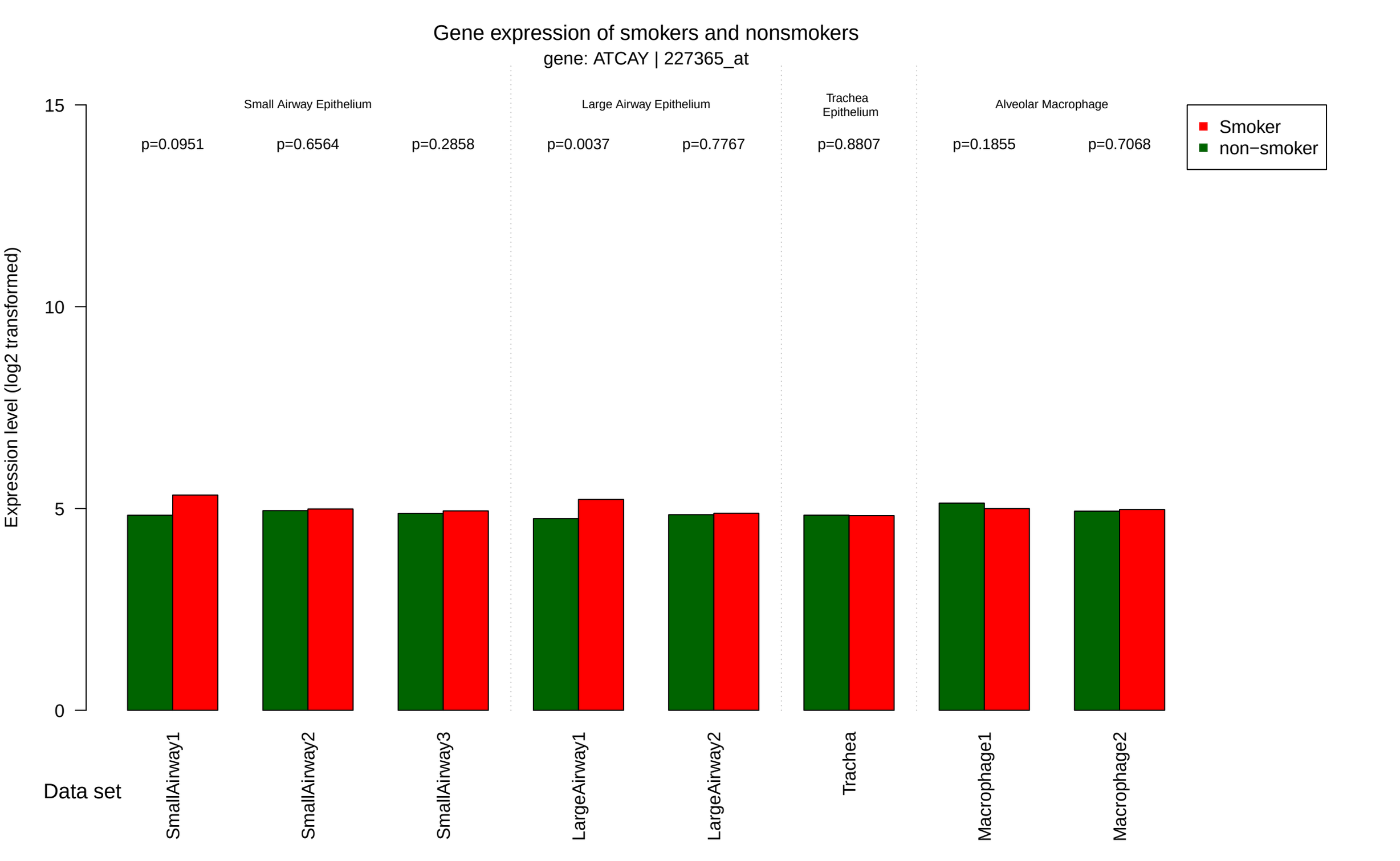
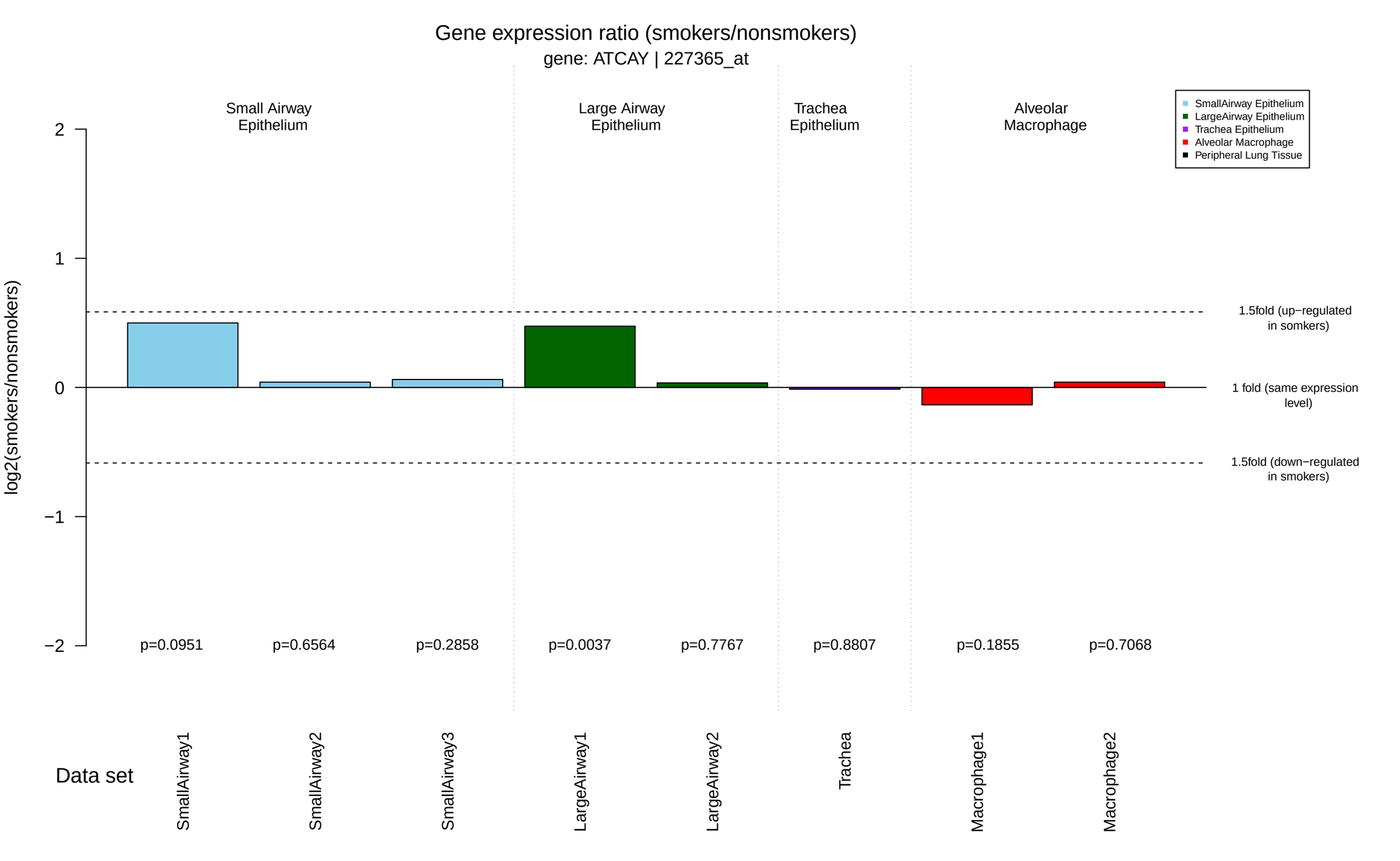
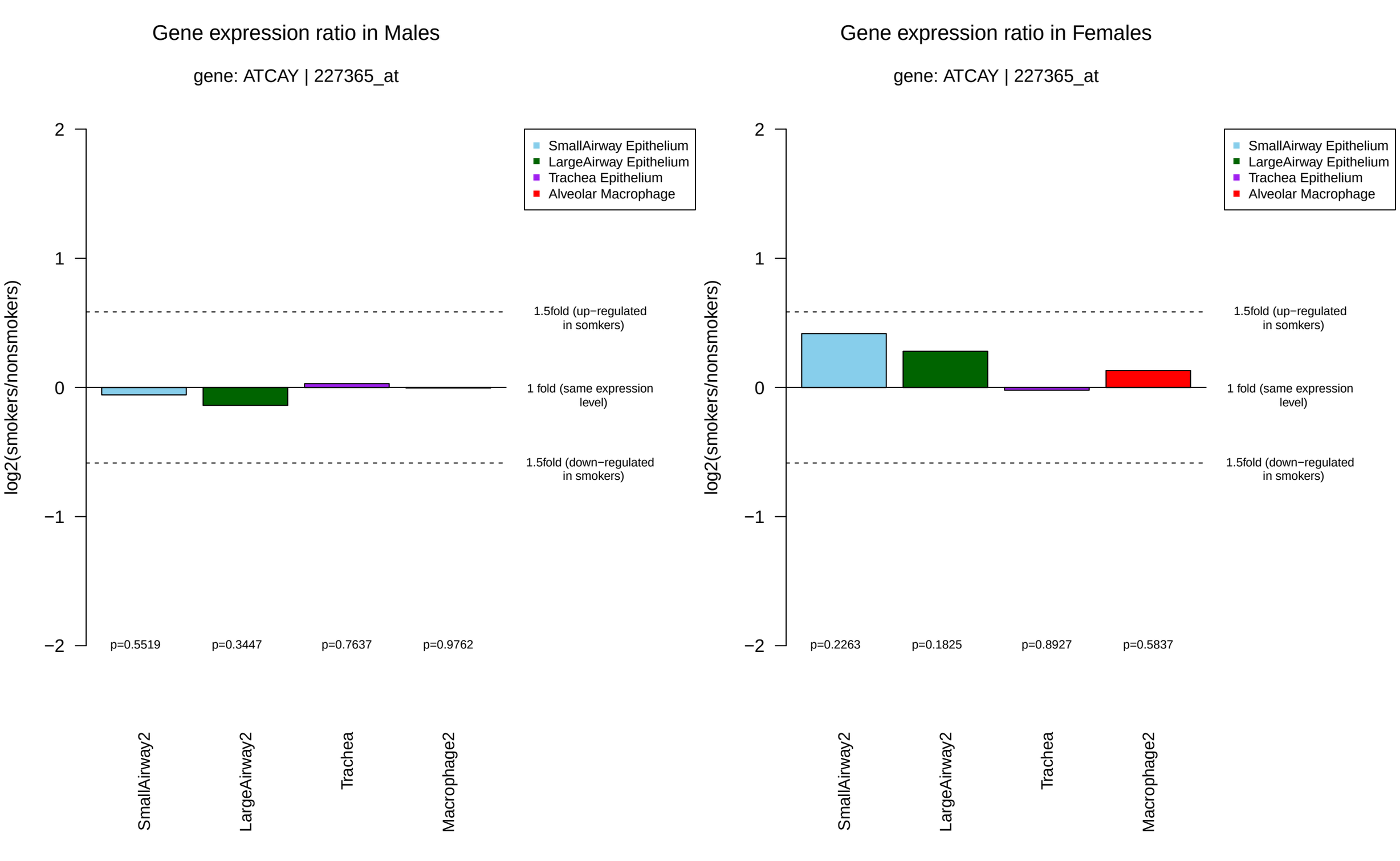
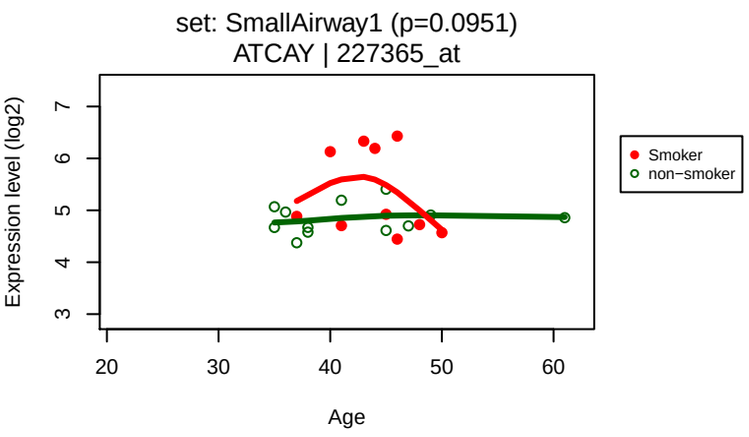
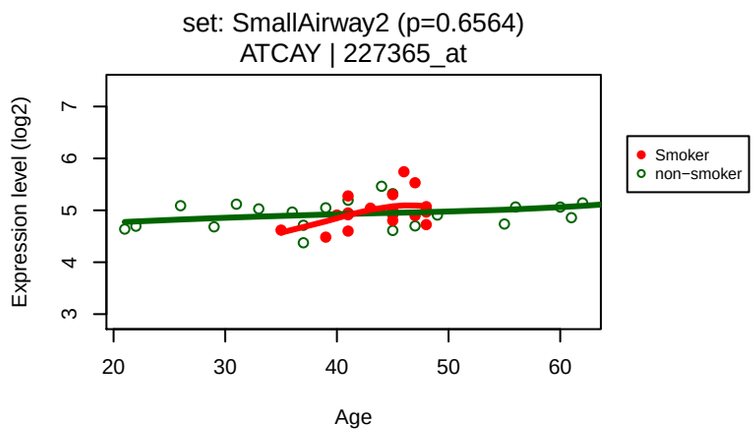
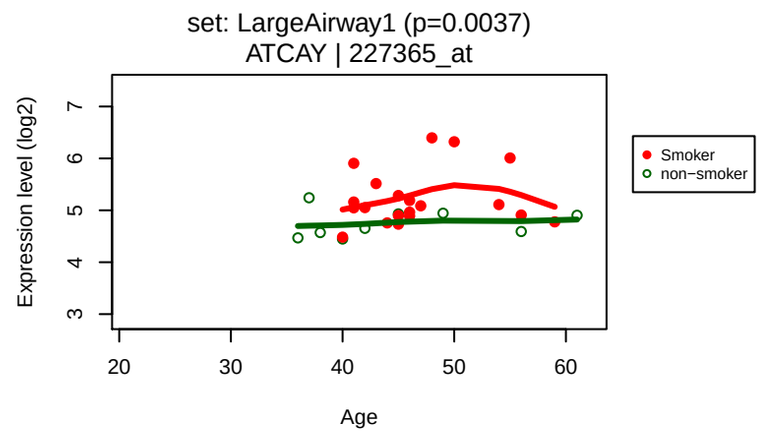

Gene expression of smokers and nonsmokers
gene: ATCAY | 227365_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0951
p=0.6564
p=0.2858
p=0.0037
p=0.7767
p=0.8807
p=0.1855
p=0.7068
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

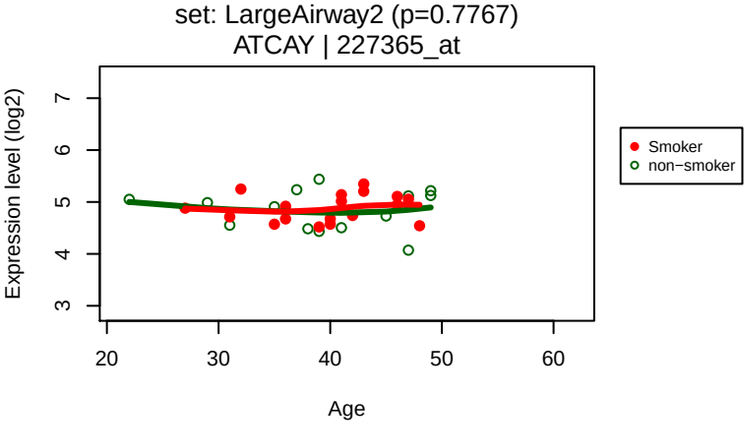

Gene expression ratio (smokers/nonsmokers)
gene: ATCAY | 227365_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0951
p=0.6564
p=0.2858
p=0.0037
p=0.7767
p=0.8807
p=0.1855
p=0.7068
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

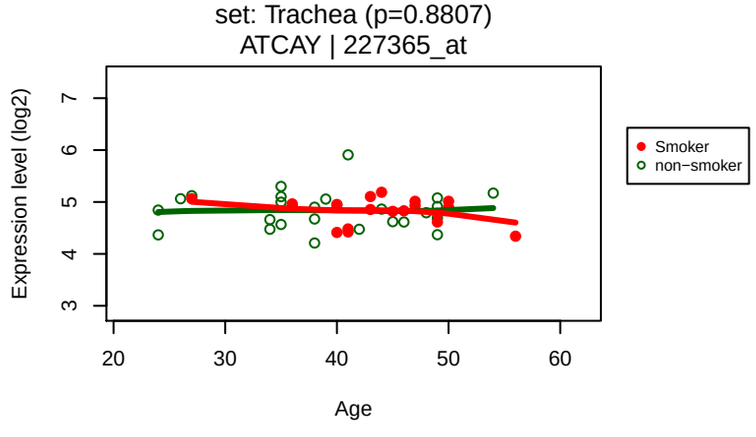

Gene expression ratio in Males
gene: ATCAY | 227365_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5519
p=0.3447
p=0.7637
p=0.9762
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATCAY | 227365_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2263
p=0.1825
p=0.8927
p=0.5837
SmallAirway2
LargeAirway2
Trachea
Macrophage2

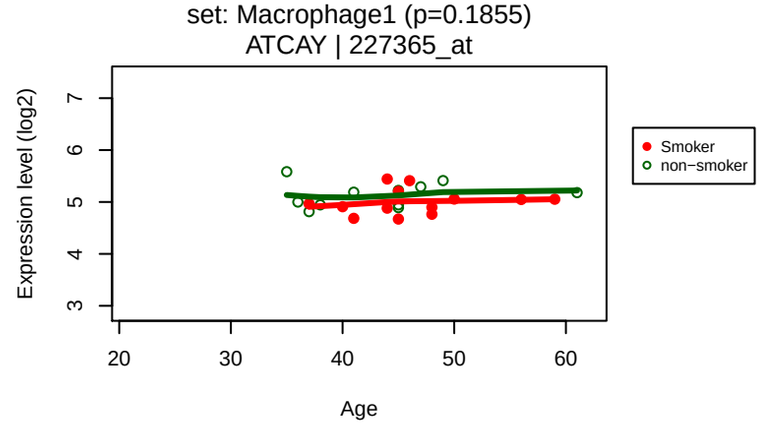

set: SmallAirway1 (p=0.0951)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker

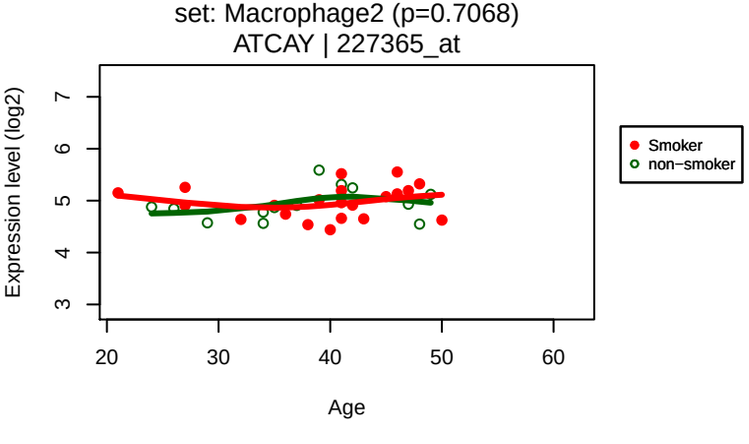

set: SmallAirway2 (p=0.6564)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker

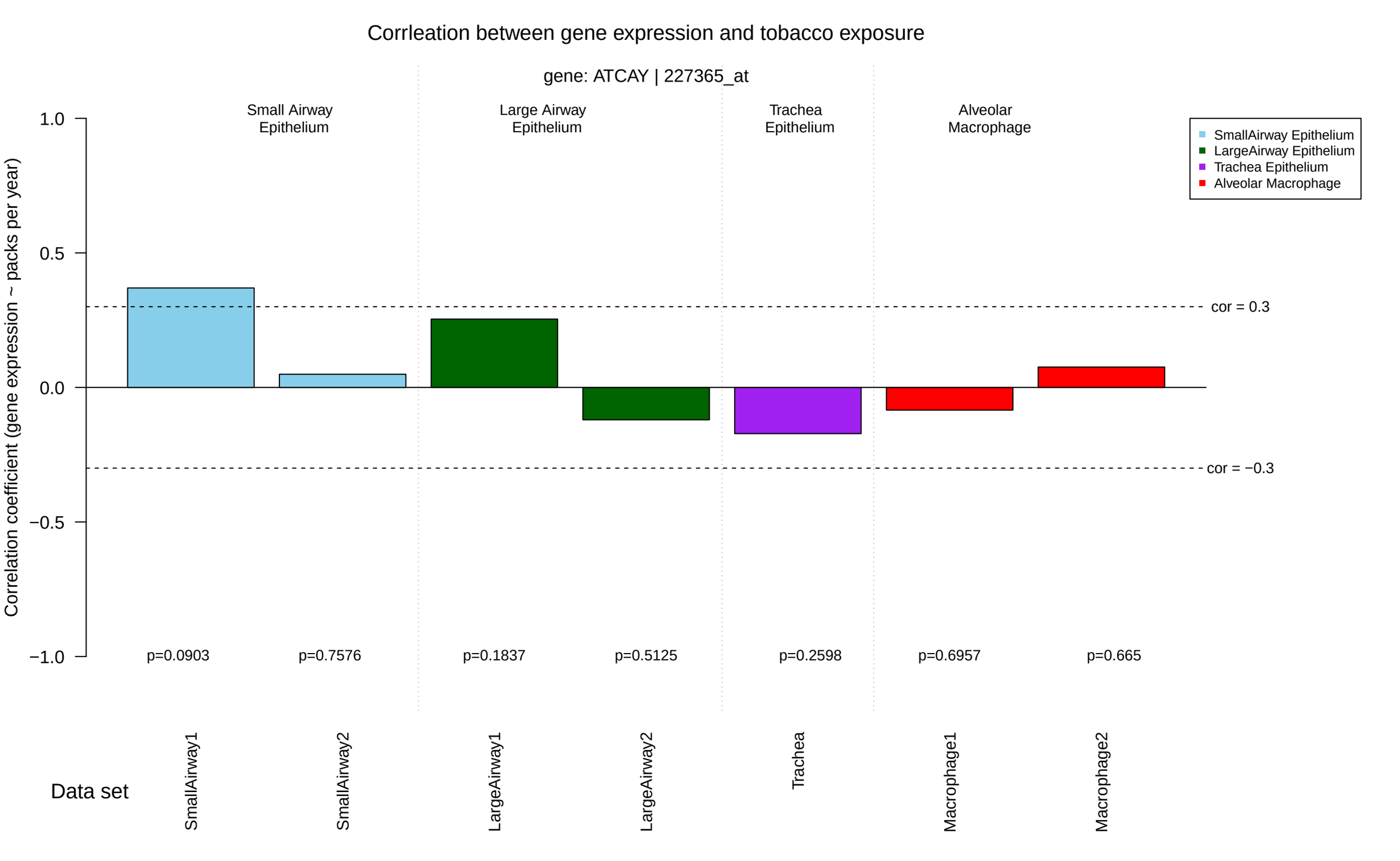

set: LargeAirway1 (p=0.0037)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7767)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8807)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1855)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7068)
ATCAY | 227365_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATCAY | 227365_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0903
p=0.7576
p=0.1837
p=0.5125
p=0.2598
p=0.6957
p=0.665
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2